Mathematical Statistics | **Exam 2018-01-12 at 8:00-13:00**
Centre for Mathematical Sciences | **MASA01 B**
Lund University | **Inference theory**

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Calculators are allowed. Only statistical tables provided by the institute are allowed. No other tables or formula sheets are allowed.

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1. Suppose that one tosses a coin repeatedly until one gets a head for the first time, and that succeeds on the 4'th try (so the first 3 tosses result in tail and the 4'th toss results in head). One suspects that the coin is flawed, meaning that the probabilities that one gets head and tail are different, and wants to test this. Make an exact test of the hypotheses

$$\begin{aligned} H_0 &: \quad \text{the coin is fair,} \\ H_1 &: \quad \text{the coin is flawed.} \end{aligned}$$

[10p]

2. The r.v. X has a probability density function

$$f(x) = \begin{cases} \frac{1}{4\theta}, & 0 \le x \le \theta, \\ \frac{3}{4}\theta e^{\theta x}, & x \le 0. \end{cases}$$

One has observations $-1.37, -2.21, -2.91, 0.11, 0.94, 0.59, 0.18, 0.29$. Find the maximum likelihood estimator of θ. [10p]

3. The log exchange rate r between two currencies A and B can be assumed to be $N(\mu, \sigma^2)$-distributed, with unknown μ and σ^2. The economy can be in a stable mode or unstable mode, with $\sigma^2 \le 0.9$ defined as a stable mode. One has observed the exchange rate during six days, and the data are $-1.46, -1.42, 0.929, 0.557, -0.728, -1.02$. The above data are assumed to be independent and assumed to be collected when the economy is in the same mode. One can formalise hypotheses

$$\begin{aligned} H_0 &: \quad \text{the economy is in a stable mode,} \\ H_1 &: \quad \text{the economy is in an unstable mode.} \end{aligned}$$

Test the hypotheses on significance level 0.05. [10p]

4. Suppose that X is $Bin(25, \theta)$ distributed and suppose that we have the observation $x = 5$.

   (a) Find the least squares estimator of θ. [5p]

   (b) Perform an exact test, on significance level 0.01, of the hypotheses

$$\begin{aligned} H_0 : \quad & \theta \geq 1/3 \\ H_1 : \quad & \theta < 1/3. \end{aligned}$$

   [5p]

   (c) Construct an approximate test for the above hypotheses on significance level 0.01. Calculate the (approximate) power of the constructed test at $\theta = 1/5$. [10p]

5. The r.v. X is $Un(0, \theta)$-distributed. An i.i.d. sample of observations of X is

$$1.97, 1.91, 1.91, 1.64, 1.59, 0.97, 1.86, 0.000242, 0.535, 0.442, 1.379, 1.404.$$

   (a) Find the maximum likelihood estimator of θ. [6p]

   (b) Construct a 95 % upper bounded confidence interval for θ. [14p]

6. The time to an event T is a common concept in survival analysis, together with the survival function of T, defined as $S(t) = P(T > t) = 1 - F_T(t)$. In an experiment with 30 individuals that were followed until they experienced the event in question one gets the following observations (unit is days) of T

$$\begin{aligned} &24, 26, 43, 24, 40, 42, 17, 35, 4, 37, 31, 43, 41, 44, 21, \\ &41, 49, 16, 44, 29, 27, 12, 19, 41, 27, 24, 41, 9, 16, 19. \end{aligned}$$

   (a) Find the plug-in estimator $\hat{S}(t)$ of $S(t)$ at the point $t = 25$. [6p]

   (b) Find a theoretical expression for the expectation $\mu(t) = E(\hat{S}(t))$ and the variance $\sigma^2(t) = Var(\hat{S}(t))$, of the estimator $\hat{S}(t)$, and give estimates of $\mu(25)$ and $\sigma^2(25)$. [6p]

   (c) Construct and perform a normal-based (approximate) test of the hypotheses

$$\begin{aligned} H_0 : \quad & S(25) \leq 1/4 \\ H_1 : \quad & S(25) > 1/4. \end{aligned}$$

   What is the power of the constructed test at $S(25) = 1/2$? [8p]